



## **Report on the Glaciological Studies Carried out during XXIII Indian Antarctic Expedition**

**Amit Dharwadkar<sup>1</sup>, R. Ravindra<sup>2</sup>, D. Jayapaul<sup>1</sup>, S. Jayaram<sup>3</sup>, P.R. Golani<sup>4</sup>  
and V. Ravikant<sup>5</sup>**

- <sup>1</sup>. Antarctica Division, Geological Survey of India, Faridabad
- <sup>2</sup>. National Centre for Antarctic and Ocean Research, Goa
- <sup>3</sup>. Geological Survey of India, Bangalore
- <sup>4</sup>. AMSE Wing, Geological Survey of India, Jaipur
- <sup>5</sup>. Geochronology Lab., CHQ, Geological Survey of India, Kolkata

### **9 DECEMBER 2003**

Antarctic Glaciers Speed Up: Scientists at NSIDC have found that glaciers around the area of the Larsen B Ice Shelf accelerated immediately after it collapsed early in 2002. (NSIDC)

### **1 DECEMBER 2003**

Ice Shelf Feels the Heat: "Warming waters may doom the Antarctic Peninsula's Larsen Ice Shelf, and other Antarctic ice shelves could be more endangered than had been thought, according to a paper published in the 31 October issue of Science." NSIDC researcher Ted Scambos is quoted in the paper. (Science Magazine)

### **6 OCTOBER 2003**

Huge Antarctic iceberg makes a big splash on sea life (EurekAlert)

### **22-24 SEPTEMBER 2003**

Giant Arctic ice shelf splits: Largest Arctic ice shelf breaks up, freeing all water from the freshwater lake it dammed. (ABC Science Online, BBC News, Belfast Telegraph, CBC North, CBS News, CNN, CTV, Discovery Channel, Edmonton Journal, Edmonton)

Above headlines illustrate that the possibility of global climate change and its effects have caught the attention of both the scientific community and the public. Several GCMs[Global Climate Model] in vogue predict that the global warming has effected a higher degree of melting of glaciers (particularly in the polar regions). This contributes to a rise in sea level changes which can have drastic effects in low lying coastal areas around the world. The polar ice-sheet is a critical component of the GCM and our understanding of the growth and movement of Antarctic ice is vital for testing the GCMs as well as predictions for the future.

This paper presents an account of glaciological studies carried out during the austral summer of 2003-04 in continuation of the work done by GSI since the 2<sup>nd</sup> Indian Antarctic Expedition. The glaciological studies carried out during the 23<sup>rd</sup> IAE can be categorized as follows:

1. Iceberg monitoring.
2. Fluctuations in the Polar Ice front.
3. Accumulation/ablation studies on ice-shelf.

## **ICEBERG MONITORING**

Icebergs have their origin in the cliffs of Antarctica's coast line and form some of the most spectacular sights in the world. These icebergs float over the sea surface and move along with the prevalent ocean currents in the southern seas. These can measure up to several hundred meters in length and present a significant hazard to navigation.

Iceberg monitoring was taken up on board the ship MV Emerald Sea during the onward and return journey to/from Antarctica. The icebergs encountered were systematically recorded using the standard format of iceberg monitoring devised by the Norwegian Polar Institute. During the onward journey, the first iceberg was sighted at Lat 44°97'S and long 12°51'E. It is generally observed that the first iceberg sighting has shifted northwards with time [Fig-1]. This may be related to the higher instance of iceberg formation in recent times. A total of 207 icebergs of various sizes and shapes were recorded between Lat 44°47'S and 69°29'S during the onward journey.

During return voyage a total of 128 icebergs were sighted between 69°36'S and 55°07'S. Fig. 2(a) shows the distribution of the icebergs with respect to latitude and size during onward journey while Fig. 2(b) shows corresponding distribution during return journey. It can be seen that during the onward journey, maximum concentration of icebergs occurs in a zone

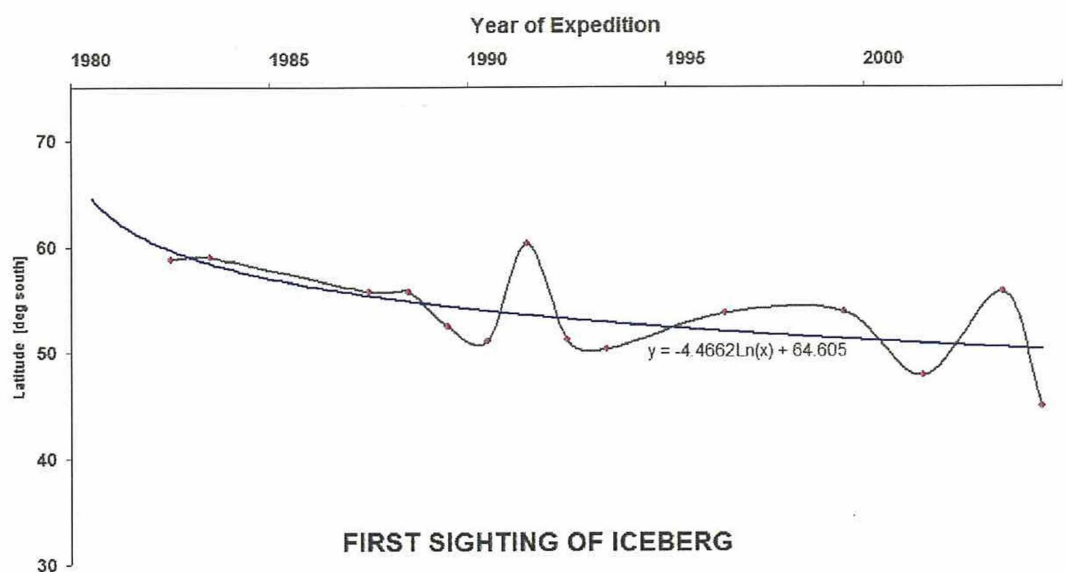


Fig. 1

Onward Voyage

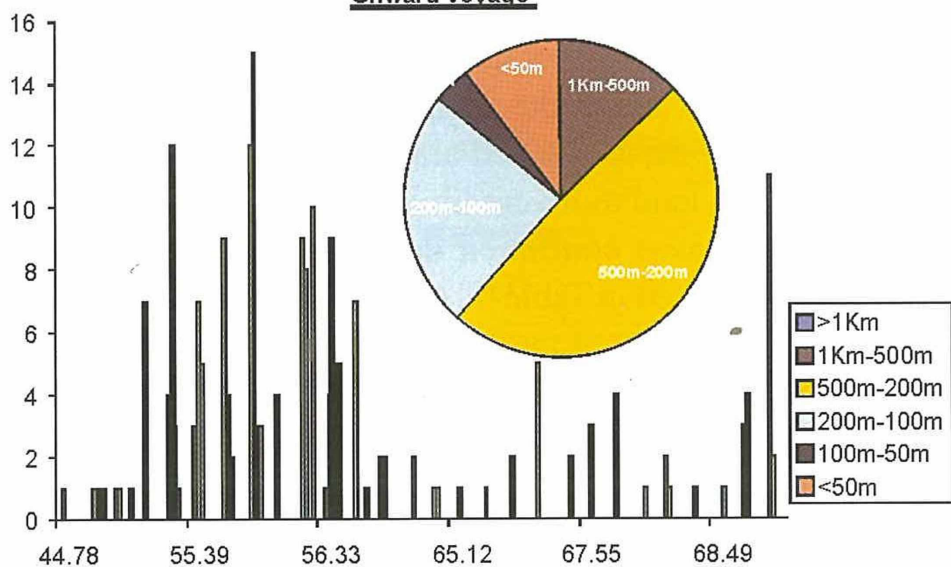


Fig. 2a

Return Voyage

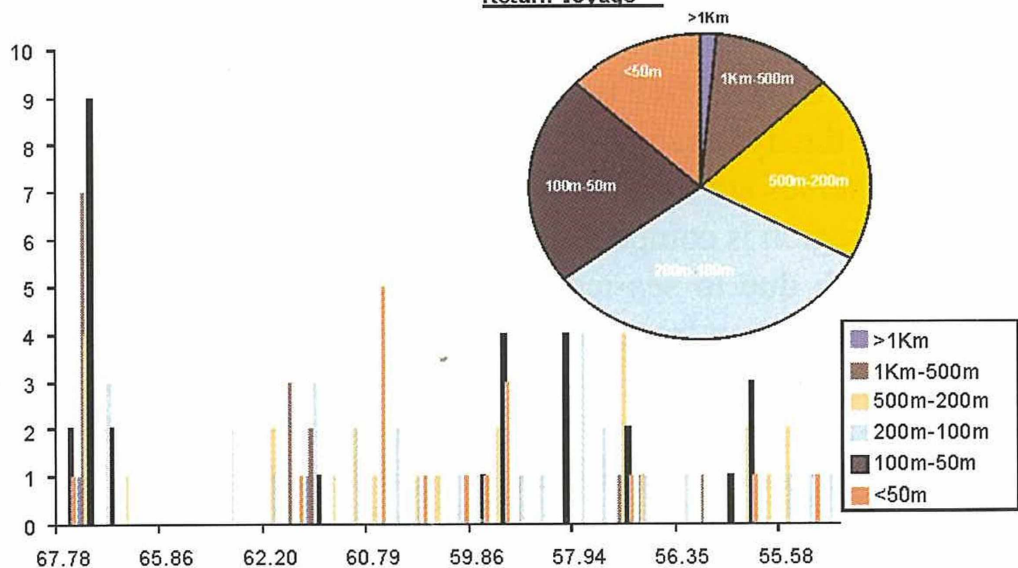


Fig. 2b

between 52°S and 59°S latitudes. During the return journey (late March 2004), the icebergs are concentrated in two distinct zones ie: 67°-66°S and 62°-55°S; with iceberg free zone in between. The size distribution of the icebergs points toward a skewed distribution where most of the icebergs sighted fall in the range of 500-100m. It is also generally observed that away from the continent, the icebergs are mostly of irregular shape and often tilted a feature resulting due to sculpturing by wind and waves. Nearer to the coast line, the icebergs are usually larger and tabular in shape.

### **FLUCTUATIONS IN THE POLAR ICE FRONT**

The snout of a glacial tongue ("Dakshin Gangotri" Glacier) overflowing the Schirmacher Oasis is being monitored by GSI since more than two decades. It has been observed that the snout is retreating at an average annual rate of 0.7m. This amounts to a 14m retreat during two decades. Since 1996, the snout is being monitored using more 'fixed' reference points marked on the bedrock along the periphery of the snout. During the 20<sup>th</sup> IAE, such points were further added along the western ice front bordering Schirmacher Oasis to lend more credibility to the observations made. The ice-front monitoring was continued during the 23<sup>rd</sup> expedition and the observations are presented in Table-1. At a few locations observations could not be taken due to collapse of the ice margin. An average annual retreat of 5.35m was recorded during 2003-04 for the entire polar margin including the DG snout. The range of this annual retreat varies between 0.38 to 14m. At two points an advance of 0.57m (XX4) and 5.27m (XX16) was observed (Figs 3, 4).

### **ACCUMULATION/ABLATION STUDIES ON ICE-SHELF**

GSI initiated accumulation/ ablation studies on the ice shelf in the India Bay region with an objective of characterizing the ice-shelf in this region and to quantify the dynamic processes occurring in this confined ice-shelf. Confined ice shelves are those which are firmly grounded to the continent. The India Bay region is comprised of a composite ice shelf in the sense that it is formed both due to sea-ice and the coalescence of glacial tongues flowing in to this region from the Polar ice-cap. Since, precipitation in the Antarctic region is very low, most of the snow accumulation in the shelf is due to aeolian snow deposition. The significance of this wind deposition can be judged by the fact that the first Indian Antarctic Station- the DG station is completely buried under a canopy of about 7-8m of snow in a span of two decades.

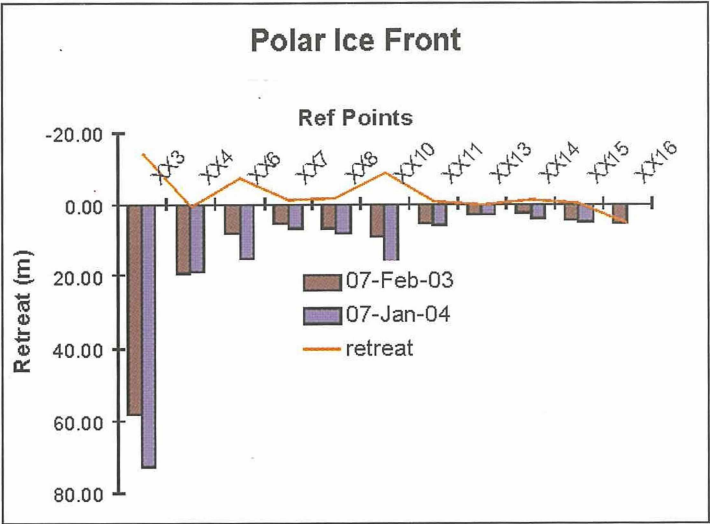


Fig. 3

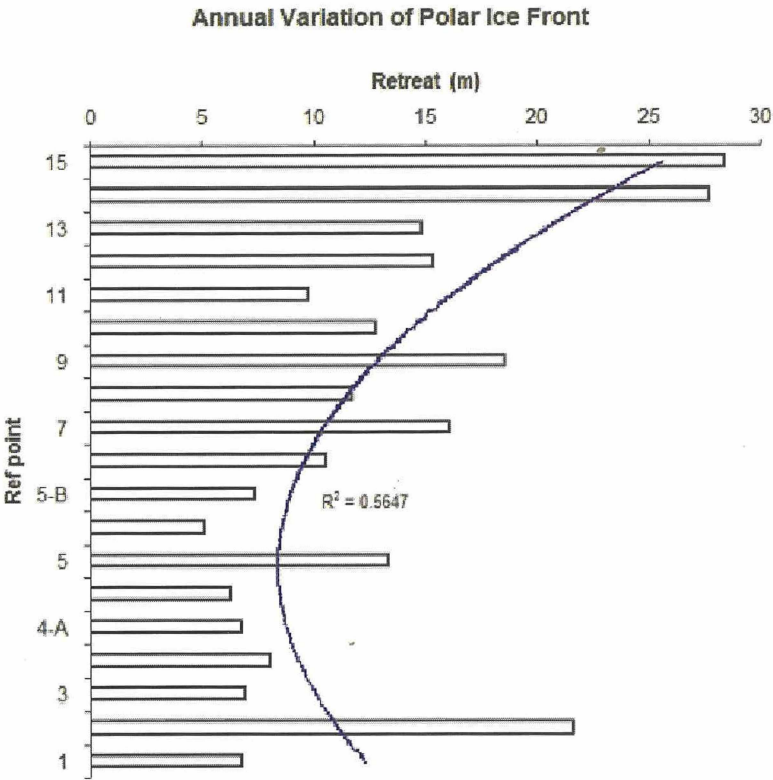


Fig. 4

**Table-1**

| DG snout data |            | Distance of ice front from obsv. point |        |        |                     |
|---------------|------------|----------------------------------------|--------|--------|---------------------|
| Point No.     | Latitude   | Longitude                              | Feb-03 | Jan-04 | Retreat/Advance (m) |
| 1             | 70°45.566' | 11°34.614'                             | 1.91   | 4.8    | 2.89                |
| 2             | 70°45.548' | 11°34.626'                             | 7.69   | 13.94  | 6.25                |
| 3             | 70°45.534' | 11°34.640'                             | 2.7    | 4.2    | 1.5                 |
| 4             | 70°45.526' | 11°34.653'                             | 2.66   | 5.33   | 2.67                |
| 4-A           | 70°45.513' | 11 °34.670'                            | 2.27   | 4.45   | 2.18                |
| 4-B           | 70°45.514' | 11°34.674'                             | 1.93   | 4.33   | 2.40                |
| 5             | 70°45.505' | 11°34.681'                             | 1.56   | 11.75  | 10.19               |
| 5-A           | 70°45.510' | 11 °34.628'                            | 1.99   | 3.08   | 1.09                |
| 5.B           | 70°45.504' | 11 °34.695'                            | 2.98   | 4.37   | 1.39                |
| 6             | 70°45.490' | 11°34.113'                             | 3.56   | 6.97   | 3.41                |
| 7             | 70°45.480' | 11°34.712'                             | 6.64   | 9.44   | 2.80                |
| 8             | 70°45.469' | 11°34.687'                             | 3.71   | 8      | 4.29                |
| 9             | 70°45.429' | 11°34.646'                             | 8.52   | 10     | 1.48                |
| 10            | 70°45.427' | 11°34.573'                             | 4.32   | 8.45   | 4.13                |
| 11            | 70°45.434' | 11°34.507'                             | 3.46   | 6.3    | 2.84                |
| 12            | 70°45.422' | 11°34.445'                             | 5.44   | 9.88   | 4.44.               |
| 13            | 70°45.428' | 11°34.374'                             | 6.99   | 7.86   | 0.87                |
| 14            | 70°45.412' | 11°34.356'                             | 11.82  | 15.9   | 4.08                |
| 15            | 70°45.406' | 11°34.289'                             | 10.28  | 18.1   | 7.82                |

## Extended Line km data

| Point No. | Latitude   | Longitude   | 07-Feb-03         | 07-Jan-04<br>not<br>observed |
|-----------|------------|-------------|-------------------|------------------------------|
| XX2       | 70°44.603' | 11 °26.390' | 44.10             |                              |
| XX3       | 70°44.621' | 11°26.529'  | 58.50             | 72.50                        |
| XX4       | 70°44.729' | 11°27.660'  | 19.22             | 18.65                        |
| XX6       | 70°44.879' | 11 °28.592' | 7.78              | 15.00                        |
| XX6A      | 70°44'53"  | 11°28' 39"  |                   | 22.50                        |
| XX7       | 70°44.936' | 11 °28.887' | 5.24              | 6.80                         |
| XX8       | 70°45.224' | 11°31.122'  | 6.52              | 8.28                         |
|           |            |             | covered.          |                              |
| XX9       | 70°45.268' | 11°31.401'  |                   | 3.00                         |
| XX10      | 70°45.307' | 11°31.634'  | 8.91              | 17.63                        |
| XX11      | 70°45.417' | 11°33.140'  | 5.27              | 5.92                         |
|           |            |             | covered<br>by ice |                              |
| XX12      | 70°45.428' | 11 °33.289' | 2.78              | wall fall                    |
| XX13      | 70°45.402' | 11 °33.485' | 2.96              | 3.10                         |
| XX14      | 70°45.396' | 11°33.612'  | 2.66              | 3.80                         |
| XX15      | 70°45.424' | 11°33.733'  | 4.42              | 4.80                         |
| XX16      | 70°45.436' | 11°34.022'  | 5.27              | 0.00                         |
|           |            |             |                   | not<br>observed              |
| XX17      | 70°45.631' | 11°34.739'  | 1.54              |                              |
| A2        | 70°45.862' | 11°35.507'  |                   | 8.91                         |

Monitoring of the surface snow accumulation on the shelf was carried out using a network of 16 stakes fixed earlier. The observations are given in Table-2. These observations show that an average accumulation of 0.52m has occurred during the year 2003-04. A graphic presentation of the

observed data (Fig 5) shows that the accumulation is not uniform ever in a small area of 0.4 sq km and varies from 0.79m to 0.25m. The data indicates that the D.G. Station area lies in a zone of accumulation on the ice shelf.

**TABLE—2**

| Stake No. | Lat        | Long       | Jan-03 | Mar-04 | net accumulation (m) |
|-----------|------------|------------|--------|--------|----------------------|
| 1         | 700431.3   | 1201 30.0  | 3.02   | 2.35   | 0.67                 |
| 2         | 700431.6   | 1201 51.5  | 3.04   | 2.64   | 0.4                  |
| 3         | 700431.9   | 120212.1   | 2.91   | 2.35   | 0.56                 |
| 4         | 70 04 32.4 | 1202 36.1  | 3.4    | 2.64   | 0.76                 |
| 5         | 70 04 27.3 | 120234.8   | 3.06   | 2.51   | 0.55                 |
| 6         | 70 04 26.8 | 120213.2   | 2.96   | 2.71   | 0.25                 |
| 7         | 70 04 26.4 | 12 01 55.8 | 3.06   | 2.57   | 0.49                 |
| 8         | 70 04 25.6 | 1201 32.1  | 2.93   | 2.58   | 0.35                 |
| 9         | 70 04 20.1 | 12 01 34.6 | 2.87   | 2.43   | 0.44                 |
| 10        | 700421.0   | 1201 54.7  | 3.06   | 2.59   | 0.47                 |
| 11        | 700421.2   | 120215.5   | 3.09   | 2.58   | 0.51                 |
| 12        | 70 04 22.3 | 120232.4   | 3.17   | 2.38   | 0.79                 |
| 13        | 700413.1   | 120234.1   | 2.97   | 2.28   | 0.69                 |
| 14        | 70 04 13.6 | 120218.4   | 3.1    | 2.56   | 0.54                 |
| 15        | 700414.9   | 1201 54.8  | 3.1    | 2.47   | 0.63                 |
| 16        | 700415.6   | 12 01 36.0 | 3.03   | 2.52   | 0.51                 |

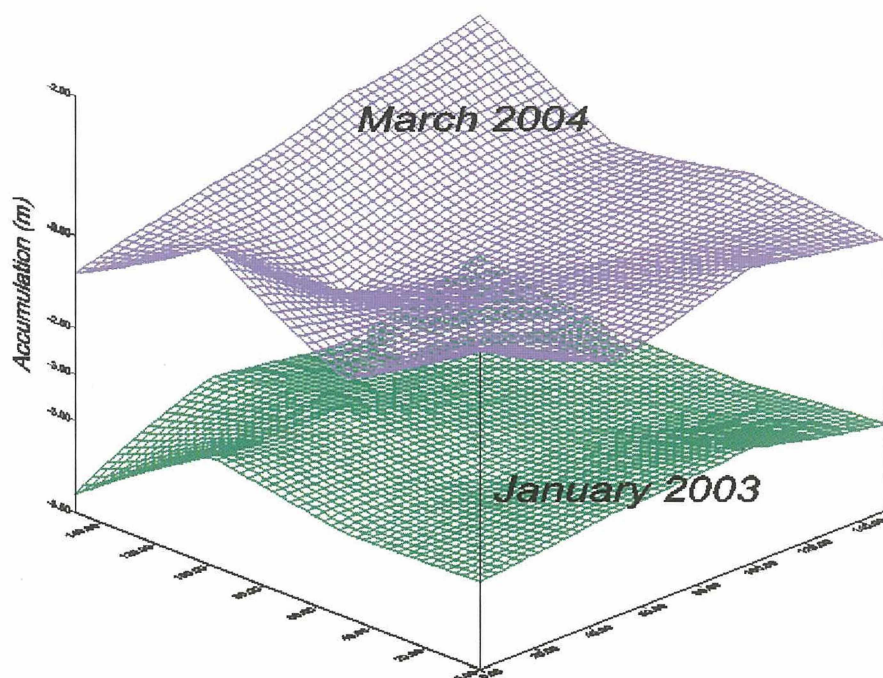


Fig. 5

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